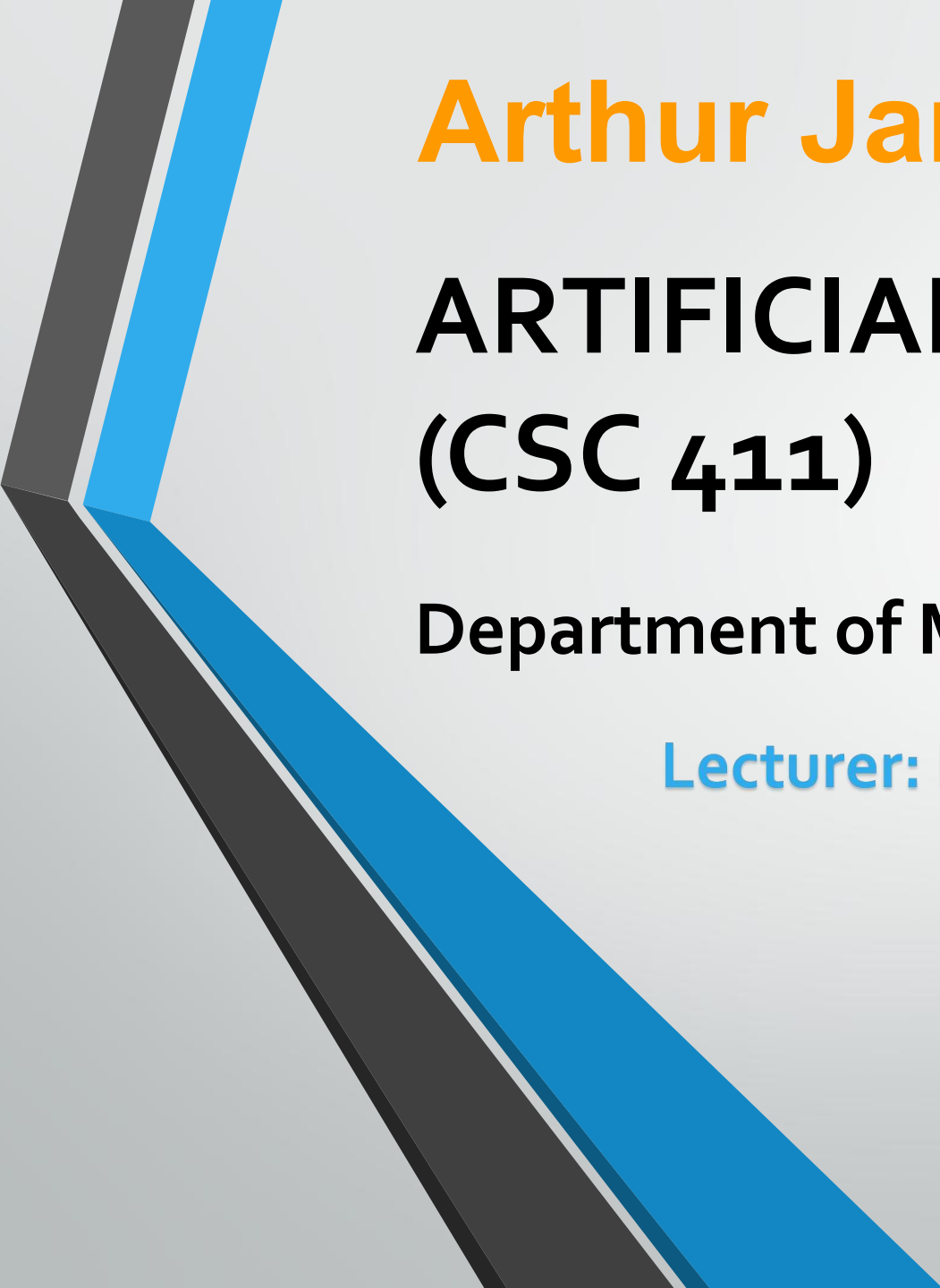




Arthur Jarvis University

**ARTIFICIAL INTELLIGENCE
(CSC 411)**

Department of Mathematics and Computer Science

Lecturer: Mrs. Sylvia A. Oladimeji

COURSE OUTLINE

- AI fundamentals, concepts, goals, types, techniques, branches, applications, AI technology and tools
- Intelligent agents, their performance, examples, faculties, environment and architectures.
- knowledge representation, semantic network and frames along with their applicable uses.
- Natural Language Processing.
- Programming languages for AI and expert systems technology, and employ application domains of AI.

Book source:

- *Artificial Intelligence: A Modern Approach, Third Edition, Stuart J. Russell and Peter Norvig*

PART 01

- AI MEANING
- AI CONCEPTS
- AI FOUNDATIONS
- GOALS OF AI
- HISTORY OF AI
- APPLICATIONS OF AI

AI MEANING

- AI is based on four approaches as shown in the image.
- It can be described as a field of study that focuses on creating intelligent machines that can perform tasks that typically require human intelligence.

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i> , in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Chamiak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i> , 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.	

AI CONCEPTS

- **Some key concepts of AI include:**
 1. Intelligence: The ability of a machine to perform tasks that typically require human intelligence.
 2. Reasoning: The ability of a machine to draw conclusions from available data
 3. Learning: The ability of a machine to improve its performance on a task overtime.
 4. knowledge representation: The way in which a machine represents and stores knowledge.

AI FOUNDATIONS

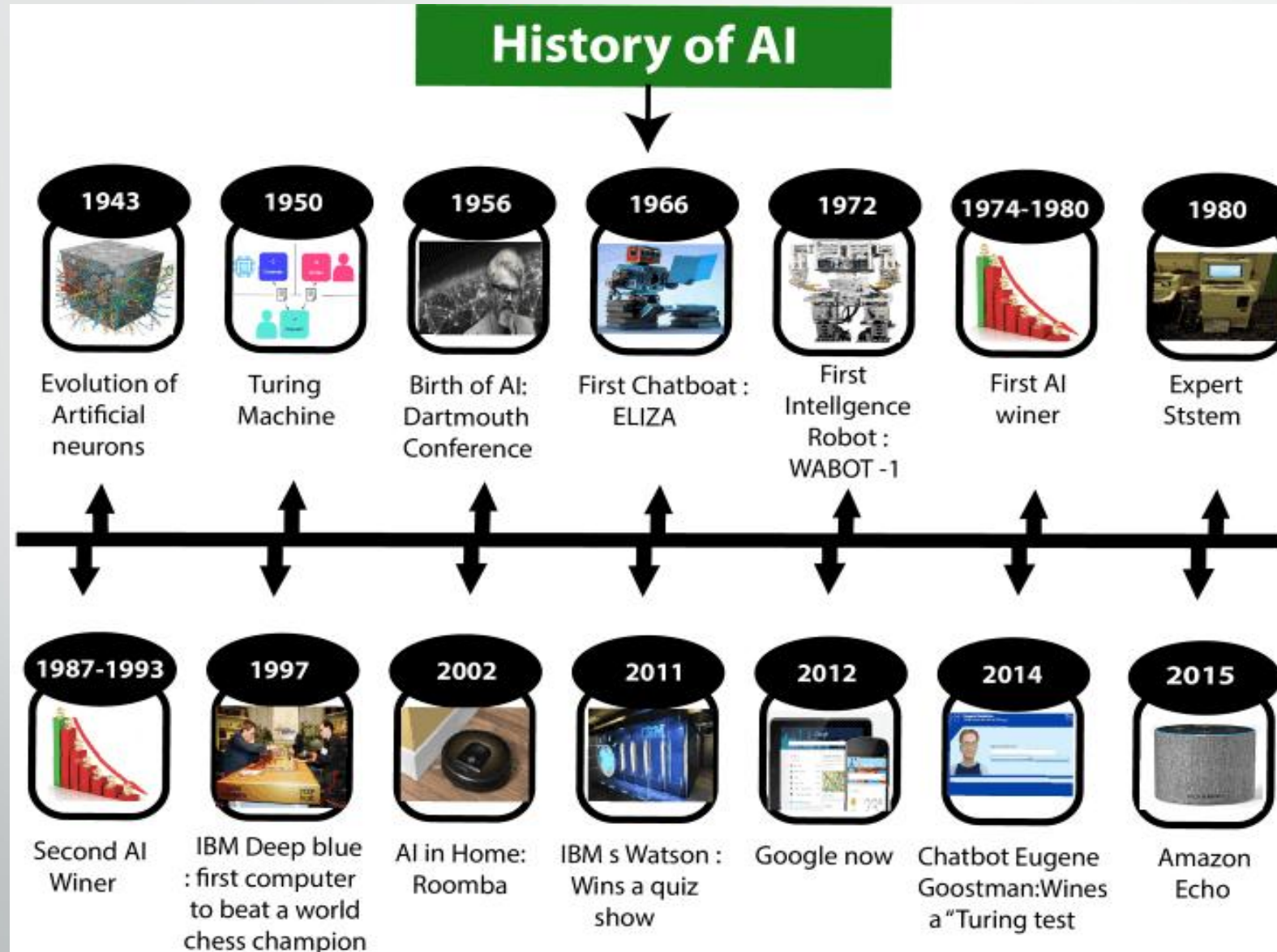
field	topics
Mathematics	1. Linear Algebra 2. Calculus 3. Probability Theory 4. Statistics
Computer Science	1. Algorithms 2. Data Structures 3. Programming Languages 4. Software Engineering
Cognitive Science	1. Cognitive Psychology 2. Neuroscience 3. Philosophy of Mind 4. Human-Computer Interaction
Knowledge Representation	1. Ontologies 2. Frames 3. Semantic Networks 4. Rule-Based Systems

GOALS OF AI

The goals of AI are:

- To create machines that can perform task that typically require human intelligence such as learning perception, reasoning, problem-solving and language understanding.
- To develop machines that can assist humans in various tasks, make decisions autonomously and improve overall efficiency and productivity.

HISTORY OF AI



APPLICATIONS OF AI

Natural Language Processing (NLP)

1. Sentiment Analysis: Analyzing text data to determine sentiment and emotions.
2. Language Translation: Developing systems that can translate languages in real-time.
3. Text Summarization: Automatically summarizing long documents into concise summaries.

Computer Vision

1. Image Classification: Developing systems that can classify images into different categories.
2. Object Detection: Detecting and recognizing objects within images and videos.
3. Image Segmentation: Segmenting images into different regions of interest.

Robotics and Autonomous Systems

1. Robotics: Developing robots that can perform tasks autonomously.
2. Autonomous Vehicles: Developing self-driving cars and drones.
3. Human-Robot Interaction: Developing systems that enable humans to interact with robots.

APPLICATIONS OF AI

Healthcare and Biomedical Engineering

1. Medical Imaging Analysis: Analyzing medical images to diagnose diseases.
2. Disease Diagnosis: Developing systems that can diagnose diseases based on symptoms and medical history.
3. Personalized Medicine: Developing personalized treatment plans based on individual characteristics.

Finance and Economics

1. Predictive Analytics: Developing systems that can predict stock prices and market trends.
2. Risk Management: Developing systems that can identify and mitigate financial risks.
3. Portfolio Optimization: Developing systems that can optimize investment portfolios.

APPLICATIONS OF AI

Education and Learning

1. Intelligent Tutoring Systems: Developing systems that can provide personalized learning experiences.
2. Learning Analytics: Analyzing educational data to improve learning outcomes.
3. Automated Grading: Developing systems that can grade assignments and exams.

Environmental Sustainability and Climate Change

1. Climate Modeling: Developing models that can predict climate change and its impacts.
2. Sustainable Resource Management: Developing systems that can optimize resource usage and reduce waste.
3. Environmental Monitoring: Developing systems that can monitor environmental pollutants and detect anomalies.



THANK YOU
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End of Week 1-2

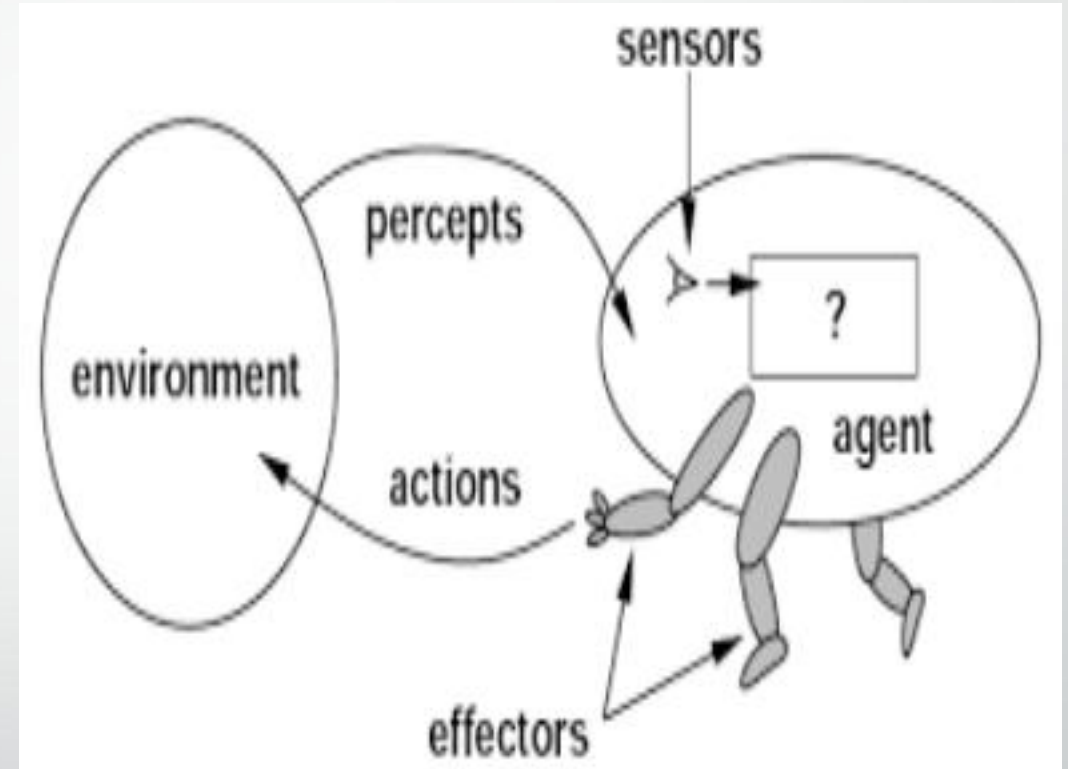
PART 02

INTELLIGENT AGENTS

- Definition of Intelligent Agents
- Considerations in intelligent agent development
- Types of intelligent agents
- Structure of agents

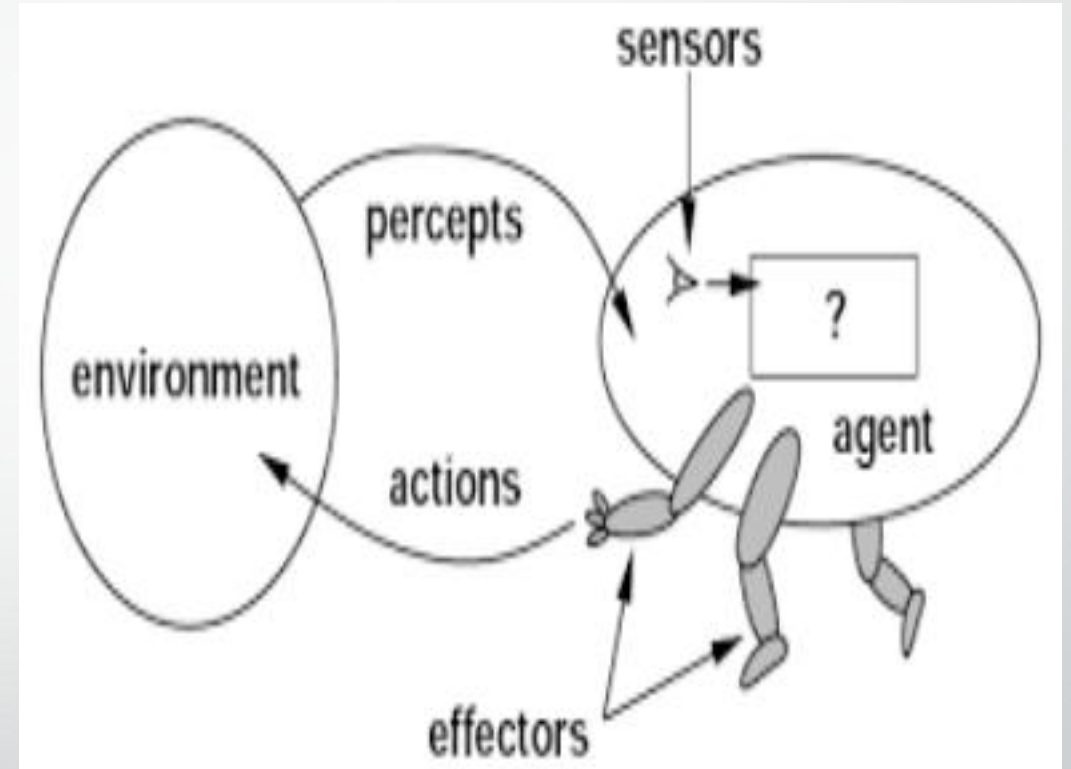
DEFENITION OF INTLLIGENT AGENT

- An Intelligent Agent is an autonomous system that **perceives** its environment, takes actions, and achieves goals using reasoning, problem-solving, and learning capabilities.
- **percept** to refer to the agent's perceptual inputs at any given instant.
- An agent's **percept sequence** is the complete history of everything the agent has ever perceived.
- The **agent function** is an abstract mathematical description of the agent's behaviour.
- The **agent program** is a concrete implementation, running within some physical system.



CONSIDERATIONS IN INTELLIGENT AGENT DEVELOPMENT

- Performance measure and rationality are usually considered when developing intelligent agents.
- A Rational Agent is an intelligent agent that acts in a rational and optimal way to achieve its goals, given its knowledge and beliefs about the environment.
- The performance measure of an intelligent agent is a metric that evaluates the agent's success in achieving its goals and objectives.
- In developing an intelligent agent, the task environment must be specified. Specifying the task environment means specifying the intelligent agent's performance measure, environment, actuators and sensors. See next slide for examples.



CONSIDERATIONS IN INTELLIGENT AGENT DEVELOPMENT

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

TYPES OF INTELLIGENT AGENTS

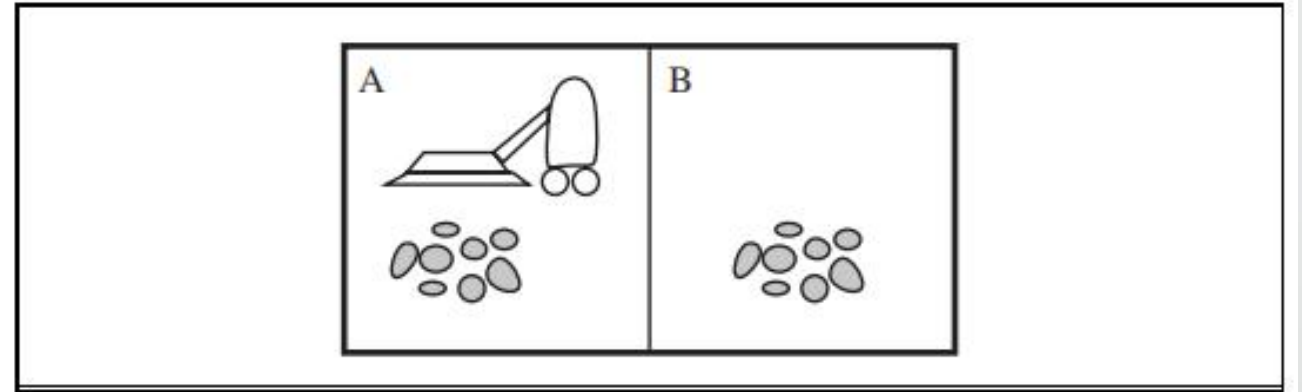
- Simple Reflex Agents: These agents react to current state without considering future consequences.
- Model-Based Reflex Agents: These agents maintain internal models of the environment to inform decisions.
- Goal-Based Agents: These type of agents pursue specific goals, using planning and decision-making.
- Utility-Based Agents: These agents make decisions based on expected utility or payoff.
- Learning Agents: These type of agents adapt and improve through experience, using machine learning algorithms.

STRUCTURE OF AGENTS

- The structure of agents is described mathematically as:
agent = architecture + agent program
- An agent program implements the agent function.
- The agent function is the mapping of percepts to actions.
- The agent program is expected to run on a computing device with physical sensors and actuators. This is called the architecture. An example is shown in the next slide.

STRUCTURE OF AGENTS

ARCHITECTURE →



AGENT
FUNCTION →

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

STRUCTURE OF AGENTS

AGENT PROGRAM



```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
               table, a table of actions, indexed by percept sequences, initially fully specified

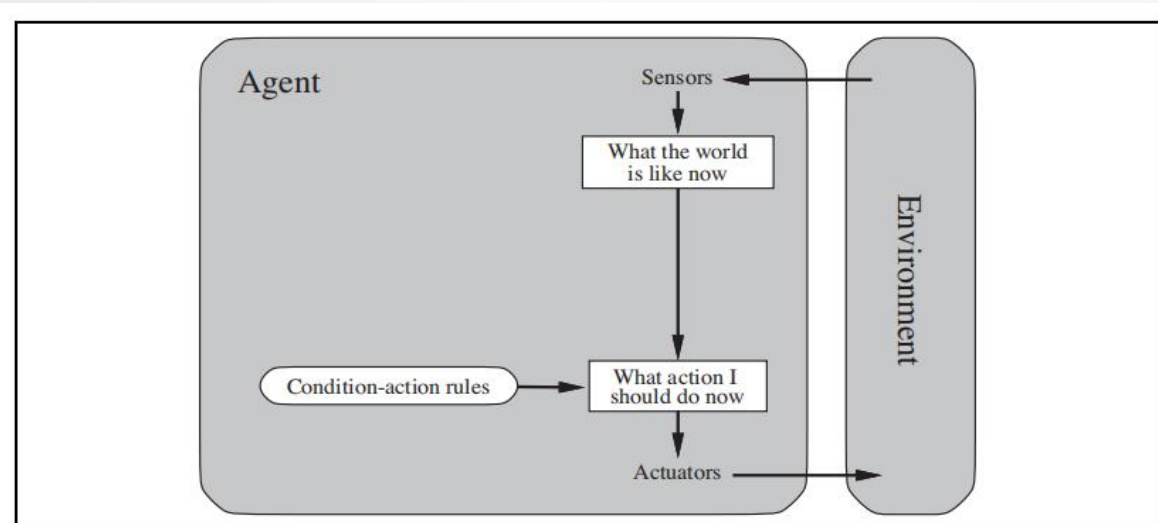
  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

STRUCTURE OF AGENTS

A SIMPLE REFLEX AGENT PROGRAM



SCHEMATIC DIAGRAM



AGENT PROGRAM

```
function SIMPLE-REFLEX-AGENT(percept) returns an action  
persistent: rules, a set of condition–action rules  
  
  state ← INTERPRET-INPUT(percept)  
  rule ← RULE-MATCH(state, rules)  
  action ← rule.ACTION  
  return action
```



THANK YOU
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End of Week 3-4

PART 03

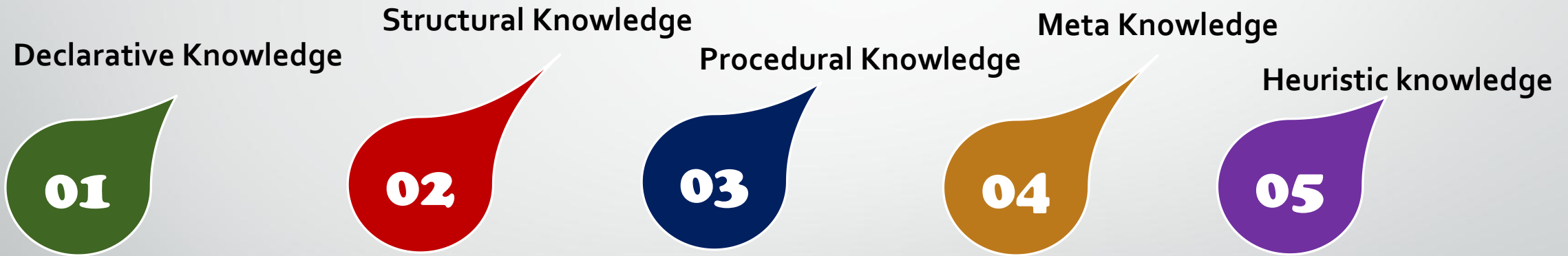
KNOWLEDGE REPRESENTATION IN AI

- What is knowledge representation
- Different types of knowledge
- Cycle of Knowledge representation
- Relation between knowledge and intelligence
- Techniques of Knowledge representation
- Representation requirements
- Approaches with example

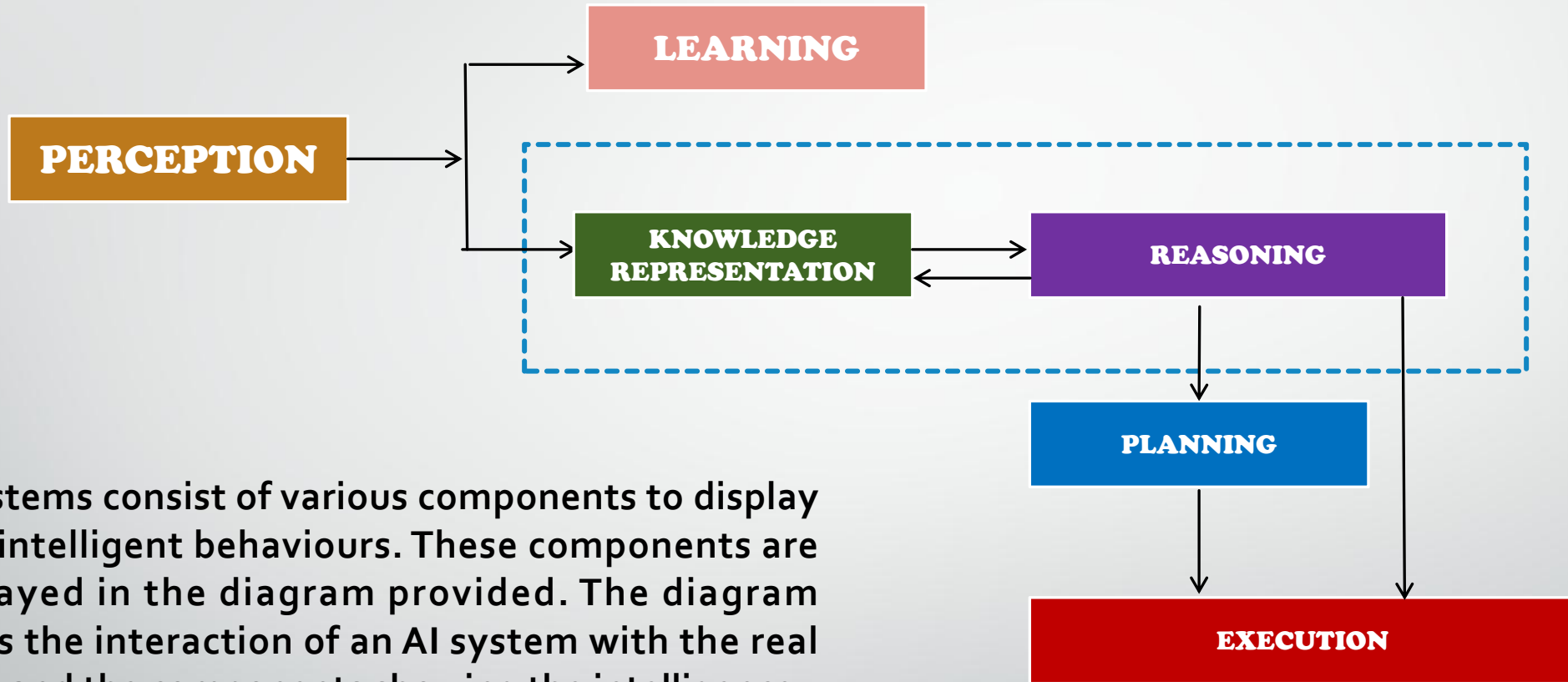
WHAT IS KNOWLEDGE REPRESENTATION?

- Knowledge representation in AI describes the representation of any knowledge. It is the study of how beliefs and judgments of any intelligent agent can be expressed suitably for automated reasoning.
- knowledge representation and reasoning represents information from the real world for a computer to understand and then utilize this knowledge to solve complex real-life problems like communicating with human beings in natural language.
- There are different kinds of knowledge that needs to be represented in AI, we have;-objects, events, performance, facts, meta-knowledge and knowledge-base.

TYPES OF KNOWLEDGE REPRESENTATION

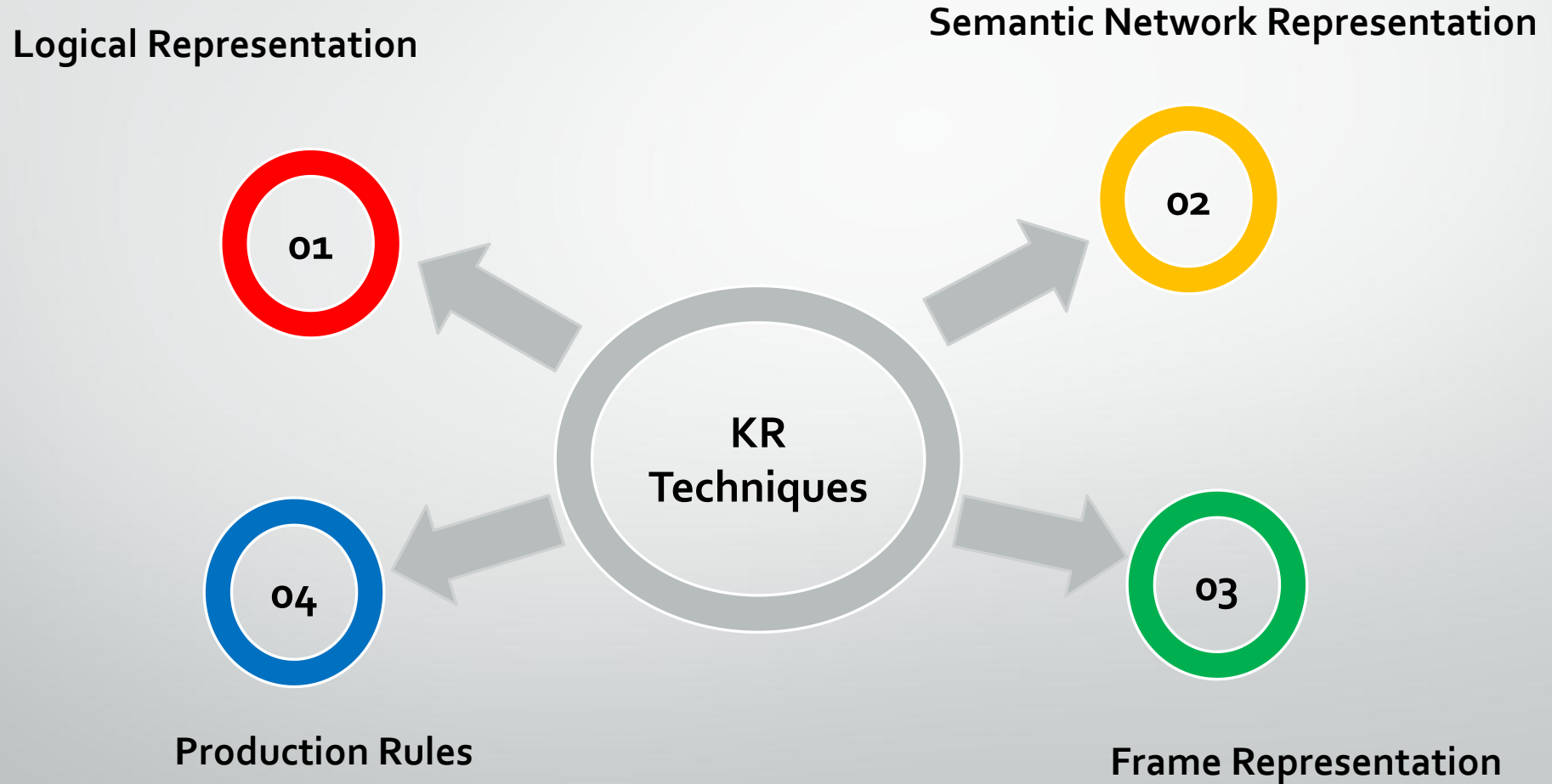


CYCLCLE OF KNOWLEDGE REPRESENTATION



AI systems consist of various components to display their intelligent behaviours. These components are displayed in the diagram provided. The diagram shows the interaction of an AI system with the real world and the components showing the intelligence.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI



TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Logical Representation

- Logical representation is a language with some **definite rules** which deal with propositions and has no ambiguity in representation. It represents a conclusion based on various conditions and lays down some important **communication rules**.
- It consist of precisely defined syntax and semantics which support the inference.

Syntax

- It decides how we can construct legal sentences in logic.
- It determines which symbol we can use in knowledge representation
- Also, how to write those symbols

Semantics

- Semantics are the rules by which we can interpret the sentence in the logic.
- It assigns a meaning to each sentence

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Logical Representation (Advantages)

- Logical representation helps to perform logical reasoning .
- The rerpresentation is the basis for the programming languages.

Logical Representation (Disadvantages)

- Logical representations have some restrictions and are challenging to work with.
- This technique may not be very natural and inference may not be very efficient.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Semantic Network Representation

- Semantic networks work as an alternative of predicate logic for knowledge representation.
- In semantic networks, you can represent your knowledge in the form of graphical networks.
- This network consists of nodes representing objects and arcs which describe the relationship between those objects.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Semantic Network Representation (Advantages)

- Semantic networks are natural representation of knowledge.
- It conveys meaning in a transparent manner.
- These networks are simple and easy to understand

Semantic Network Representation (Disadvantages)

- Semantic networks take more computational time at runtime.
- These networks are not intelligent and depend on the creator of the system.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Frame Representation

- A frame is a record-like structure that consists of a collection of attributes and values to describe an entity in the world.
- These are the AI data structures that divide knowledge into sub-structures by representing stereotypes and situations.
- It consists of a collection of slots and slot values of any type and size. Slots have names and values which are called facets.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Frame Representation (Advantages)

- It makes the programming easier by grouping the related data.
- Frame representation is easy to understand and visualize.
- It is very easy to add slots for new attributes and relations.
- It is easy to include default data and search for missing values

Frame Representation (Disadvantages)

- In frame system inference, the mechanism cannot be easily processed.
- The inference mechanism cannot be smoothly proceeded by frame representation.
- It has a very generalized approach.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Production Rule Representation

- In production rules, agent checks for the condition and if the condition exists, then production rule fires and corresponding action is carried out.
- The production rules consist of a set of rules, a working memory and a recognized act-cycle.

TECHNIQUES OF REPRESENTING KNOWLEDGE IN AI

Production Rule Representation (Advantages)

- The production rules are expressed in natural language.
- The production rules are highly modular and can be easily removed or modified.

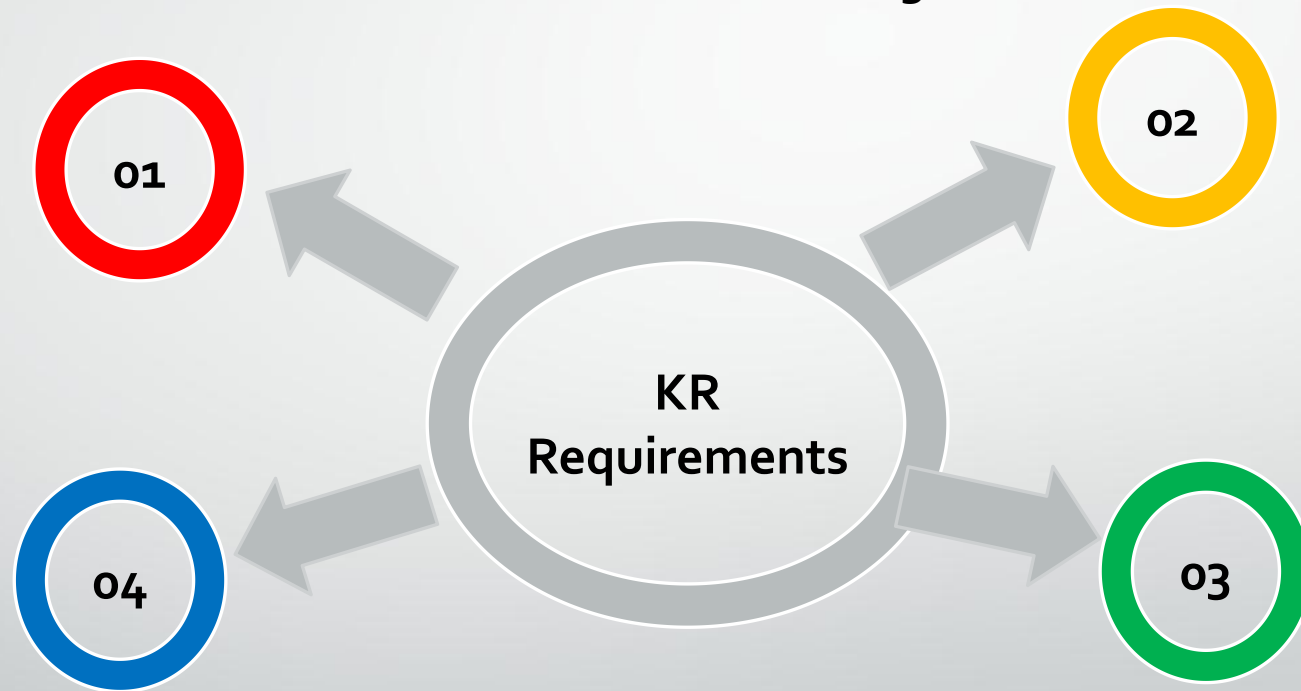
Production Rule Representation (Disadvantages)

- It does not exhibit any learning capabilities and does not share the result of the problem for future uses.
- During the execution of the program, many rules may be active. Thus rule-based production systems are inefficient.

KNOWLEDGE REPRESENTATION REQUIREMENTS

Representational accuracy: All kinds of required knowledge should be represented

Inferential adequacy: It should be able to manipulate the representation of structures to produce new knowledge corresponding to the existing structure.



Inferential Efficiency: The ability to direct the inferential knowledge mechanism into the most productive direction by storing appropriate types.

Aquisitional Efficiency: The ability to acquire new knowledge easily using automatic methods.

APPROACHES OF KNOWLEDGE REPRESENTATION WITH EXAMPLES

Simple Relational Knowledge

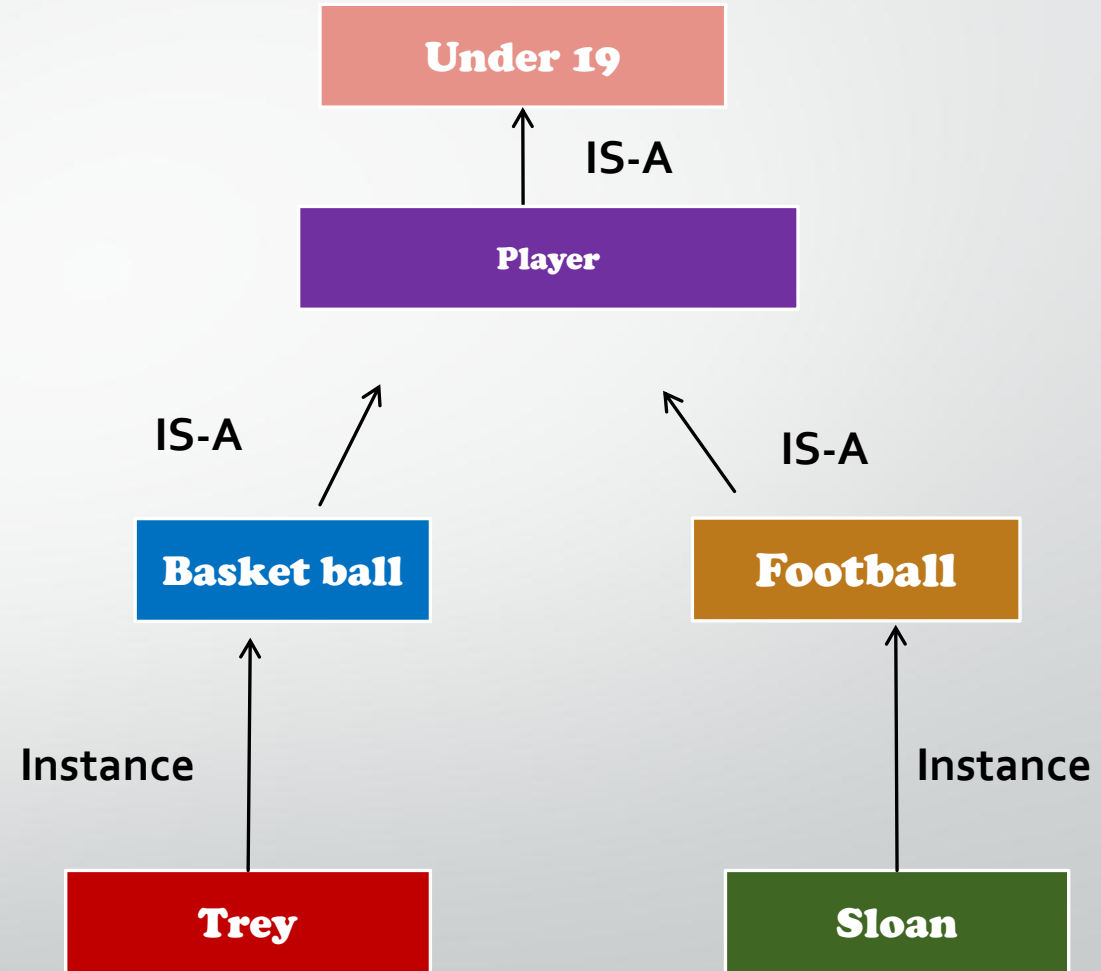
- It is the simplest way of storing facts which uses the relational method.
- Here all the facts about the sets of the objects are set out systematically in columns.
- This approach is famous in database systems where the relationship between different entities is represented.

Name	Age	EmpID
Ellis	25	000101
Amelia	22	000302
Suzzy	28	000008

APPROACHES OF KNOWLEDGE REPRESENTATION WITH EXAMPLES

Inheritable Knowledge

- In this approach, all data must be stored into hierarchy of classes and should be arranged in a generalized form in a hierarchical manner.
- This approach contains inheritable knowledge which shows a relation between instances and classes. It is called the **instance relation**



APPROACHES OF KNOWLEDGE REPRESENTATION WITH EXAMPLES

Inferential Knowledge

- The inferential knowledge represents knowledge in the form of formal logic.
- It can be used to derive more facts as well as guarantees correctness.

STATEMENT o1	Trey is a basket ball player
STATEMENT o2	All basket ball players are athletes

BasketBall_player(Trey)

$\forall x = \text{BasketBall_player}(x) \rightarrow \text{Athlete}(x)$



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End of Week 5-7

PART 04

NATURAL LANGUAGE PROCESSING (coming soon)

- Language models
- Text classification
- Information retrieval
- Information extraction



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End of Week 8-10